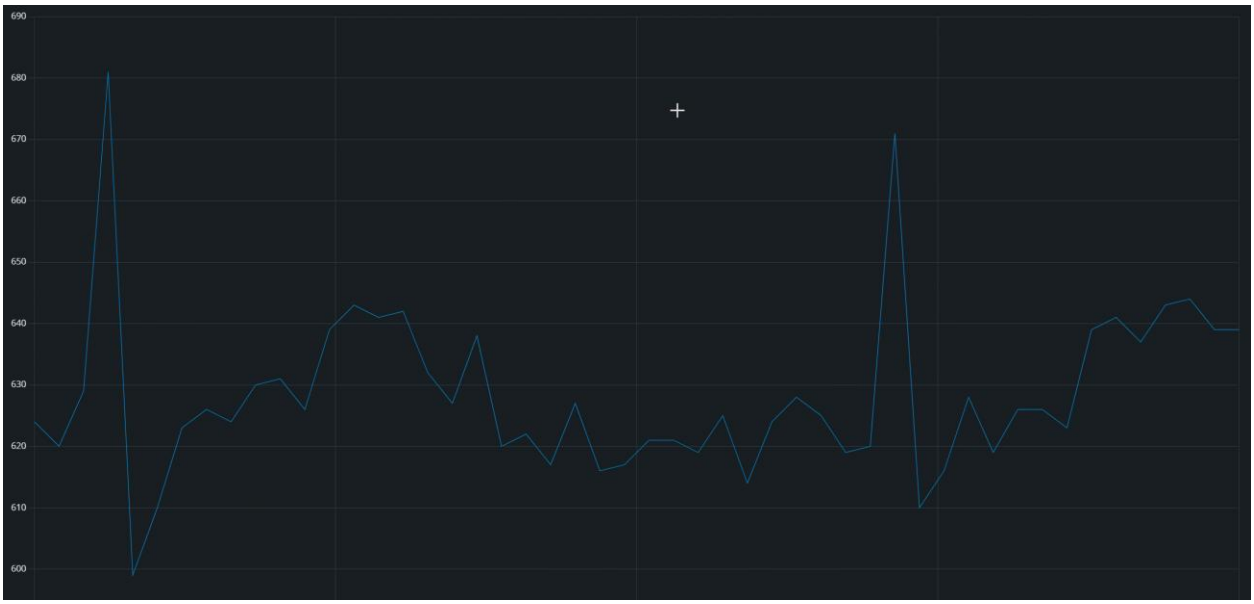
Matthew Averill

Part II-A

1. 

   The waveform is fairly noisy. Changing the leads had a significant impact on the quality of the data. The shape of the peaks generally look the same when compared to the morphology of the waveform shown in the instructions. The only difference is some noise in between the different peaks.

2. Yes, the low-pass filter was enough to reduce the noise to a relatively clean ECG. Originally I designed for a 25 Hz low-pass filter, but that resulted in too much noise. Increasing to a cutoff frequency of 35 Hz helped a lot. To further improve the signal, I could implement a digital filter within the Arduino script itself. This could help reduce the noisy peaks and “round them off”. A specific digital threshold for peaks could be set.

3.

This signal is much more noisy than the previous resistor value waveform. There are many erroneous peaks in between each beat.

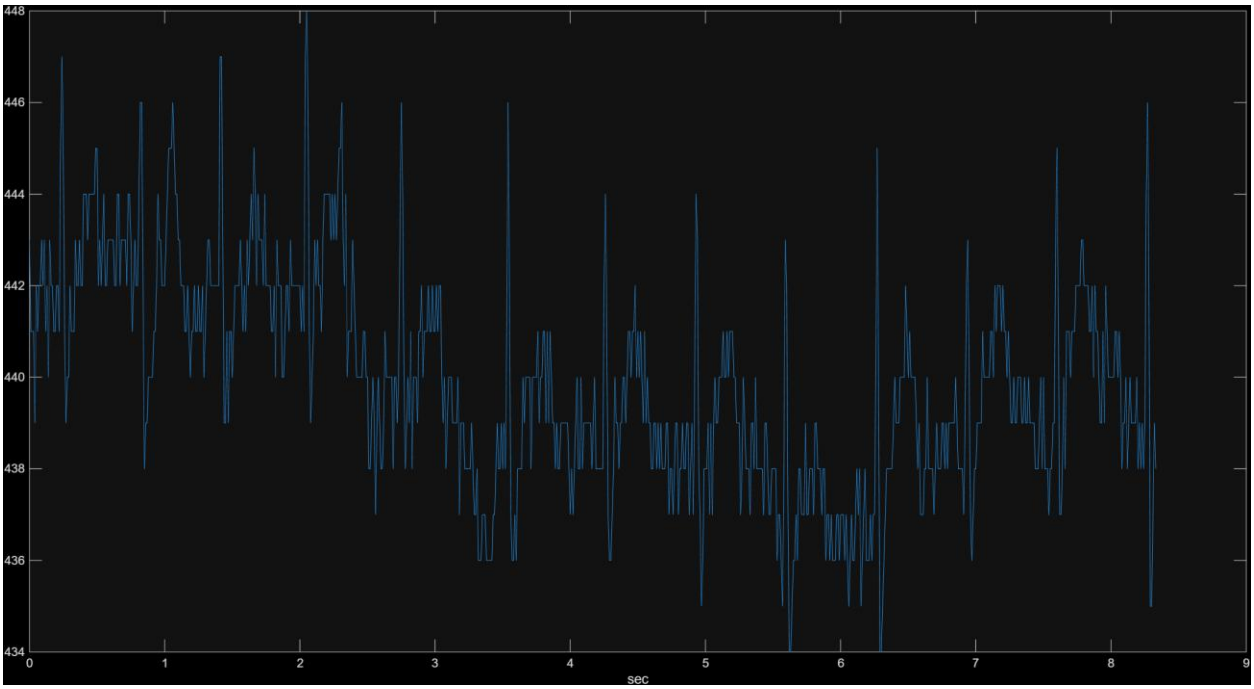
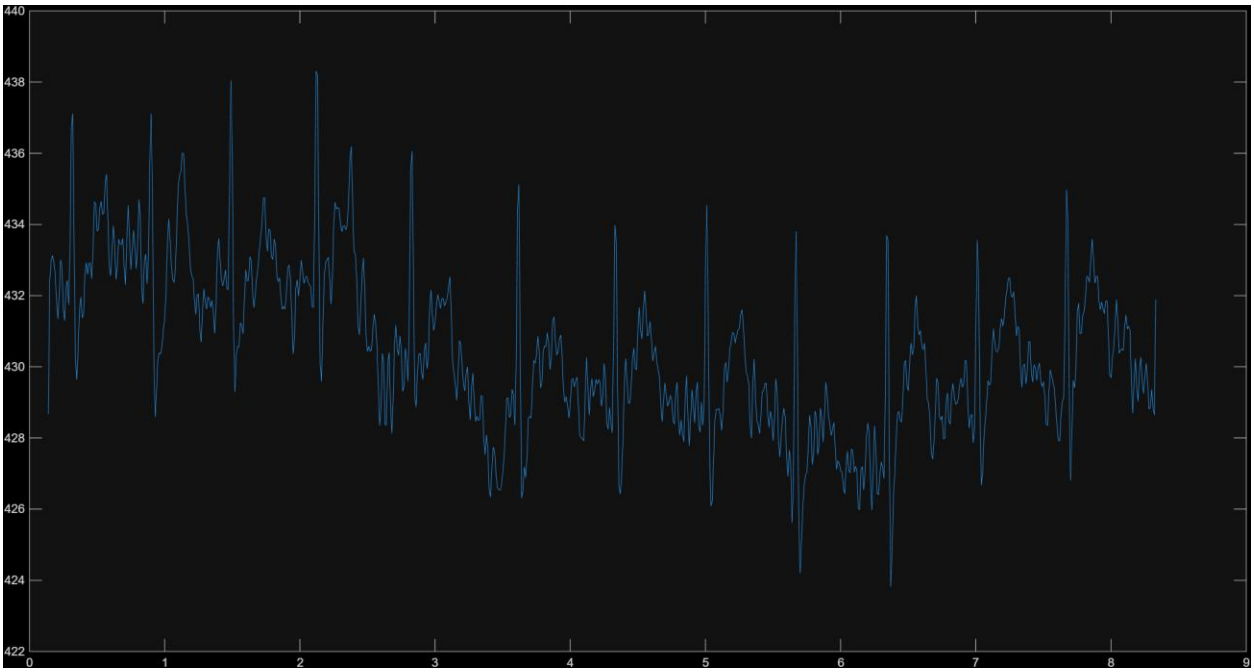
4.

When flexing my arms the waveform would act sporadically. It seems as though the noise amplitude would increase as I flex my arms or move around.

Matthew Averill

Part II-B

Raw ECG signal using MATLAB



ECG signal with a MATLAB digital low pass filter